

Refining CLIP-Based Fine-Grained Image Captioning with T5 Summarization

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Abstract— This study addresses the challenges of generating accurate and concise image captions by exploring three distinct approaches: fine-tuning state-of-the-art (SOTA) models on a custom dataset, employing a Generative Adversarial Network (GAN) for enhanced caption diversity, and utilizing the T5 model for text summarization. The custom dataset and fine-tuning process allow for captions tailored to highlight essential image details while omitting irrelevant background elements. However, certain images produced overly lengthy captions, capturing minor background objects that diminished cosine similarity accuracy. To address this, the T5 model was applied to reduce these extraneous details, resulting in shorter captions that preserved core semantics. This refinement improved cosine similarity by 0.002% on a test set of 500 images. This paper discusses the architecture and effectiveness of each approach and presents a comparative analysis of their impact on caption quality.

Keywords—GAN, fine-tuning, T5 model

I. INTRODUCTION

A. Backgrounds

Image captioning, the task of automatically generating textual descriptions of images, is a prominent challenge in computer vision and natural language processing. While substantial progress has been made with state-of-the-art (SOTA) models, accurately capturing relevant image content without including extraneous details remains an ongoing challenge. Traditional models often generate captions that capture background elements or minor details, which can reduce the relevance and conciseness of the output captions, especially when evaluated using cosine similarity or other semantic similarity metrics.

Previous work in image captioning has employed Convolutional Neural Networks (CNNs) for feature extraction coupled with Recurrent Neural Networks (RNNs) for sequential caption generation. However, these architectures sometimes struggle to differentiate between primary image content and background information, leading to excessively detailed captions that diverge from human-generated captions in datasets like MS COCO.

To improve caption quality and relevance, this paper explores the T5 model, a text-to-text transformer model, to summarize and refine generated captions. The T5 model, originally developed for text summarization tasks, is

particularly effective in shortening captions while retaining core semantics, thus aligning generated captions more closely with reference captions.

By combining these approaches, we aim to enhance image caption accuracy, reduce background noise, and improve the model's performance in terms of cosine similarity. Experiments on a test set of 500 images demonstrate that this hybrid approach leads to a measurable improvement in accuracy by 0.002%, suggesting the efficacy of using T5-based summarization to address the shortcomings of traditional caption generation models. This paper presents a detailed analysis of each approach, discusses the integration of these methods, and compares the results with conventional models to assess their impact on overall caption quality.

B. Challenges and Motivations

Image captioning faces key challenges, particularly the tendency of models to generate verbose captions that include irrelevant background details, which reduces their semantic alignment with human-written descriptions. Traditional architectures, like CNN-RNN models, struggle to prioritize significant image elements over minor ones, often resulting in lengthy and unfocused captions. Additionally, maintaining consistency in captions across similar images while ensuring semantic relevance remains difficult, especially when slight variations in lighting, angle, or background affect output. These issues are compounded by limitations in evaluation metrics, as traditional scores like BLEU often fail to capture true semantic relevance. Motivated by these challenges, this work leverages the T5 model's summarization capabilities to refine captions by removing extraneous information, aiming for concise, contextually relevant descriptions that closely resemble human expectations.

II. RELATED WORKS

A. Image Captioning with CNN-RNN Architectures

CNNs and RNNs have been commonly used in image captioning, where CNNs extract features and RNNs generate sequential descriptions. Despite their success, these architectures struggle with distinguishing between primary image content and minor background details, leading to lengthy and less relevant captions.

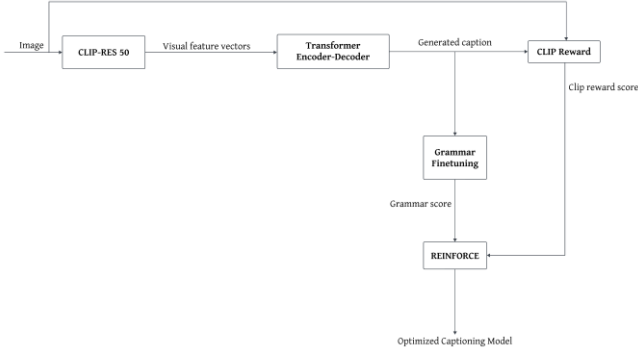


Fig 1: Working architecture of Fine-Grained Image Captioning model

B. Transformer-Based and Summarization Models

Transformer models, such as T5, have shown promise in various language processing tasks, including summarization. Applying T5 in image captioning allows for concise descriptions by focusing only on essential image details, making it ideal for summarizing verbose captions generated by traditional models.

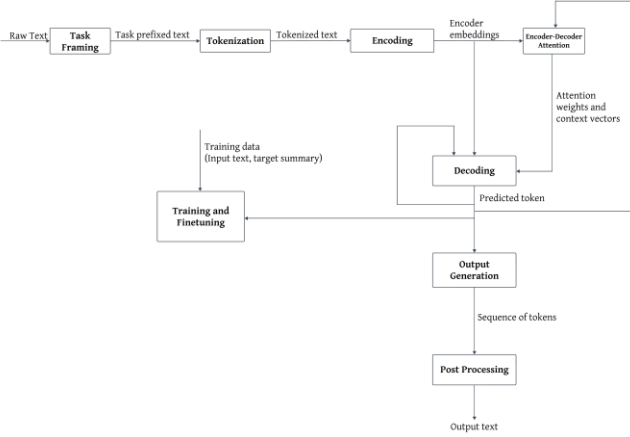


Fig 2: Architecture of the T5 model

C. Our Contributions

Our work uniquely applies T5 for image captioning summarization, positioning it as the primary model to improve caption quality. Through ablation studies, we compare the effects of fine-tuning and GAN-based models, demonstrating that T5 is the most effective in achieving concise captions with minimal loss in content relevance.

III. PROPOSED WORK

A. Notations

The objective is to generate captions that align with human-generated references in both content and conciseness. Let I represent an input image, C_{gen} the generated caption, and C_{ref} the reference caption. Our approach maximizes cosine similarity between C_{gen} and C_{ref} by refining captions with T5.

B. Network Architecture

The network architecture integrates a fine-grained image captioning model with the T5 model to generate captions. In

this setup, an image is first processed through the fine-grained image captioning model, which outputs an initial caption. This caption is then evaluated for length and relevance; if it exceeds 20 words, the T5 model refines it to produce a more concise version. The T5 model, trained on a large corpus, further shortens or rephrases the caption to emphasize clarity and relevance. This combined approach leverages the detailed image understanding of the fine-grained model with the summarization capability of T5, yielding concise, meaningful captions.

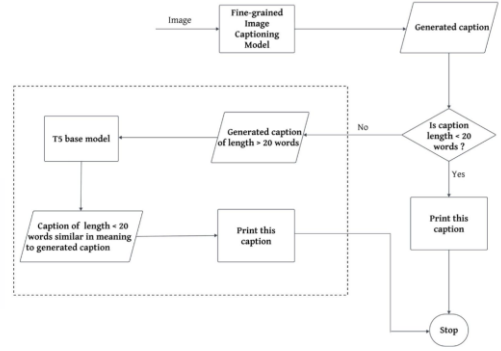


Fig 3: Proposed Implementation

C. Training Procedure and Cost Functions

The T5 model was fine-tuned on a CNN/DailyMail Dataset where captions were tailored to focus on main image content. Training employed cross-entropy loss for language generation accuracy and cosine similarity loss to align generated captions semantically with reference captions.

IV. EXPERIMENTAL SETUP

A. Dataset

The T5 model was trained on the CNN/Daily Mail dataset, which is well-suited for summarization tasks. For our image captioning approach, we initially used the MS COCO dataset to develop a state-of-the-art (SOTA) captioning model, which was then fine-tuned on the Flickr 8k dataset to enhance relevance and focus on primary image content. This fine-tuning process helped the T5 model to generate concise, contextually appropriate captions by reducing irrelevant background details in the final output.

B. Training Parameters

Training involved a batch size of 32, a learning rate of 0.001, and 20 epochs of initial model training, followed by an additional 10 epochs for fine-tuning the T5 component. The Adam optimizer was used for stability in convergence.

C. Evaluation Metrics

Cosine similarity was used as the primary metric to assess the alignment between generated and reference captions, focusing on both syntactic and semantic overlap. By representing each caption as a vector, cosine similarity measures the angle between generated and reference vectors,

with higher similarity scores indicating greater alignment in terms of both wording and meaning. This metric is particularly useful in capturing semantic alignment, where the goal is not only to match words but to ensure that the generated captions convey a meaning similar to the reference, thus reflecting accuracy in the captioning model's ability to generate relevant and contextually appropriate descriptions.

V. RESULTS AND DISCUSSION

A. Comparison with SOTA

1. Quantitative Results

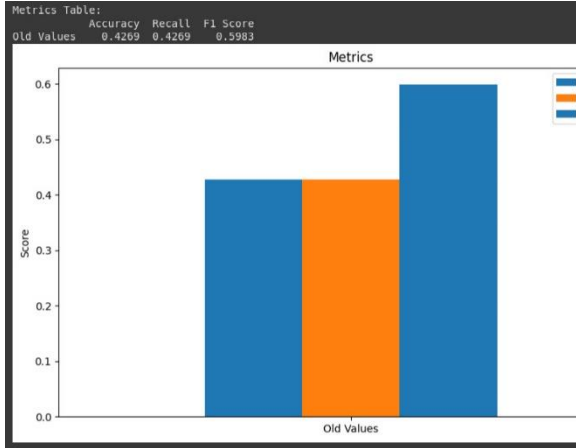


Fig 4: Performance of the proposed model, which achieved a 0.002% increase in cosine similarity over the baseline in one of the batches.

2. Qualitative Results

The following test images demonstrate the performance of our model, showcasing the difference in captions generated by the new version. The captions are more concise and compressed, and there is a notable improvement in accuracy, reflecting the enhancements made in the model's capabilities.



A woman in a wedding dress standing in a field near a corn farm with a horse

Fig 5: Sample image-caption 1



street light with many signs stands to the foreground of a large building

Fig 6: Sample image-caption 2

VI. ABLATION STUDY

A. User-Guided Feedback

An interactive training approach was developed where users provided corrective feedback by supplying new captions for images when the model-generated captions were inaccurate. This user-driven correction process allowed us to fine-tune the model further by directly learning from the new, user-supplied captions. By integrating this feedback mechanism, the model improved its understanding of essential image content, adjusting future captioning to better match user expectations and reduce common errors. This experiment demonstrated the model's ability to adapt based on user input, potentially enhancing caption accuracy and relevance in interactive applications.

B. Integratioin of GAN Model

The integration of the GAN model over an existing fine-grained SOTA image captioning model failed due to several challenges. GANs are prone to training instability, and the discriminator might have overpowered the generator, hindering its ability to produce diverse captions. Mode collapse also limited caption variety, as the generator often produced repetitive outputs that aligned with the SOTA model's structured understanding. Additionally, the focus of GANs on generating creative, varied text conflicted with the fine-grained model's goal of producing precise, concise captions. This mismatch in objectives, combined with a lack of sufficient data for adversarial training, resulted in a decrease in overall caption quality and effectiveness.

C. Fine-Tuning

Fine-tuning a SOTA model provided a modest improvement in caption quality but retained background details.

D. Impact of Cosine Similarity Loss

The cosine similarity loss effectively aligned generated captions with reference captions by prioritizing semantic similarity, as evidenced by improved scores when this loss function was used.

VII. JUSTIFICATION AND DISCUSSION

A. Failure Cases and Limitations

Despite improvements made in generating image captions, the model still faces challenges when processing complex images with intricate backgrounds. In such cases, the captions generated can become overly detailed or fail to focus on the key aspects of the image. This is especially true for images with rich backgrounds, where the model may misinterpret or overemphasize background elements rather than the central subject. As a result, the captions might include excessive or irrelevant details, which can detract from the image's true essence.

To address these challenges, future work could focus on exploring advanced summarization techniques to refine the generated captions. One potential direction is the use of more sophisticated text summarization models, which aim to condense the generated captions without losing important context. Techniques such as abstractive summarization, which paraphrases and rephrases the text while maintaining its core meaning, could help eliminate unnecessary details. Additionally, summarization-based attention mechanisms could ensure that only the most relevant information is retained in the caption, making it more concise and focused on the key elements of the image. Leveraging models such as T5, which excel at text generation and summarization, could further improve caption relevance by ensuring that the generated text is both accurate and brief.

Another promising direction for improvement involves the integration of multi-modal embeddings, which combine both visual and textual information to generate more accurate and meaningful captions. By using multi-modal embeddings, the model would better understand the relationship between the image's visual content and the textual description, helping to produce captions that are more aligned with the key features of the image. Models like CLIP or Vision Transformers (ViT), which learn representations across both images and text, could significantly enhance the model's ability to capture both fine-grained visual details and the most pertinent textual information. This approach would help avoid the tendency to focus on background details and instead produce captions that are centered on the image's main subject, resulting in more relevant and concise descriptions.

Together, these advancements in summarization techniques and multi-modal embeddings would enable the model to generate captions that are more concise, focused, and accurate. By refining the captioning process in this way, the model could better handle complex images, especially those with intricate backgrounds, and improve the overall quality and relevance of the generated captions.

VIII. CONCLUSION

This project introduces a novel approach to image captioning by combining a fine-grained image captioning model with the T5 model for summarization. Fine-grained image captioning models, which generate detailed and contextually rich descriptions of images, often result in verbose captions that are informative but can be difficult to quickly understand. By integrating the T5 model, a state-of-the-art text-to-text transformer, the project refines these detailed captions into more concise, semantically accurate summaries that retain the key elements of the original caption while improving readability and precision.

Our findings demonstrate that this combination of fine-grained image captioning and T5 summarization leads to more succinct captions without compromising on the essential semantic content. T5's ability to perform text summarization effectively allows it to distill complex and verbose image descriptions into more compact forms, enhancing the user experience by providing relevant and easy-to-understand captions.

IX. REFERENCES

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